Admittance-Controlled Internal Grasping of Hollow Objects: A Grasp Matrix Approach in Isaac Sim

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***Abstract*—Internal grasping of hollow objects provides a crucial alternative to external grasping in densely cluttered environments, but it relies strictly on force closure, requiring precise application of outward radial forces to prevent object slippage or structural damage. Because pure position control is often inadequate for navigating the narrow margin of error between slippage and damage, advanced adaptive control methodologies are necessary. This report presents a comprehensive framework combining a grasp matrix mathematical approach to analytically compute the required lifting forces with an admittance control scheme for a two-finger parallel robotic gripper. The system operates on a Hybrid Reactive-Deliberative Architecture, utilizing estimated interaction forces to dynamically adjust the gripper's position without relying on explicit force sensors, while incorporating reactive reflexes for slip-correction and instability avoidance. The methodology is validated through simulated pick-and-place experiments in NVIDIA Isaac Sim, testing the gripper on rectangular and circular hollow objects of varying masses and material friction coefficients. Results demonstrate that grasp success is highly dependent on object mass rather than material composition, while also highlighting simulation-specific challenges related to numerical instabilities when the physics engine processes complex circular collision meshes.**

***Index Terms*—Hollow object, grasp matrix, cognitive approach, admittance control, NVIDIA Isaac Sim.**

I. Introduction

Hollow objects are widespread in both daily life and industrial applications, frequently featuring circular cavities (e.g., bottles, pipes, and cans) or rectangular openings. Humans instinctively employ internal grasping strategies to manipulate such objects. This approach offers significant advantages, such as enabling the simultaneous transfer of multiple objects in a single hand [1].

In the context of robotic manipulation, internal grasping is particularly advantageous in highly cluttered environments. When objects are densely packed, such as in bin-picking scenarios or on a crowded tray, traditional external grasping is often infeasible due to the lack of clearance between adjacent items. In these situations, taking advantage of the internal cavity of the object provides a highly effective alternative.

However, a primary challenge associated with internal grasping is that it relies strictly on force closure rather than form closure. Because the robotic end-effector cannot physically enclose the object, executing a stable grasp requires the precise computation and application of internal radial forces. If the applied force is too low, the object will slip; if it is too high, the object risks structural damage or deformation.

This report addresses these challenges by presenting a comprehensive control and mathematical framework for the internal grasping of hollow objects. First, a mathematical approach utilizing the grasp matrix is introduced to analytically determine the precise force required to securely lift the object. Second, an admittance control scheme for the robotic gripper is designed to ensure adaptive, safe, and precise force application. Finally, the proposed methodology is validated through a series of simulated experimental scenarios, and the results of these simulations are discussed in detail.

II. Research problem

The core objective of this research is to achieve safe and stable internal grasping of hollow objects. This task presents several distinct challenges, primarily because internal grasping relies strictly on force closure rather than form closure. The specific components of this research problem are defined as follows:

- **Spatial limitations in clutter:** In densely packed environments, such as bin-picking tasks, traditional external grasping is often rendered impossible due to a lack of clearance between adjacent items. Exploiting the internal cavity of a hollow object offers a critical alternative to bypass these environmental constraints.
- **Absence of form closure:** Unlike external grasping mechanisms, the robotic end-effector cannot physically enclose the object. Instead, grasping stability must be achieved entirely through the outward expansion of the gripper against the inner walls of the cavity.
- **The force problem:** Because stability is entirely dependent on internal radial forces, computing and applying the precise amount of force is essential. An insufficient force application results in slippage and task failure, whereas excessive force risks causing structural deformation or catastrophic damage to the object.
- **Limitations of position control:** Navigating the narrow margin of error between slippage and structural damage requires high precision. When manipulating delicate objects or managing dimensional uncertainties, pure

position-based control strategies prove inadequate, requiring the integration of adaptive force-tracking methodologies.

III. Pick and place benchmarking

This chapter provides a review of the state-of-the-art benchmarking methodologies established within the field of robotic manipulation and pick and place.

1) **A Bin-Picking Benchmark for Systematic Evaluation of Robotic Pick-and-Place Systems (2020)**[2]: this benchmark is the most general of the three. It was designed for a bin-picking, grocery-style pick-and-place setting and evaluates not just whether an object is picked and placed successfully, but also how efficiently the system performs and where failures happen along the pipeline. Its metrics include success rate, picks per hour, cycle time, and phase-level failure breakdowns, which makes it useful for comparing complete robotic systems and identifying specific bottlenecks in perception, grasping, transport, or placement.
2) **Benchmarking Cluttered Robot Pick-and-Place Manipulation with the Box and Blocks Test (2019)**[3]: This benchmark takes a different approach by adapting the clinical *Box and Blocks* test into a robotic benchmark. Instead of focusing mainly on pipeline introspection, it emphasizes standardized task execution and comparability. The authors define three protocols, Modified-BBT, Targeted-BBT, and Standard-BBT, that gradually increase in difficulty and let researchers evaluate different aspects of manipulation performance in a controlled way. A major feature of this benchmark is its use of task time and human norms as reference points, which makes it especially useful when the goal is to compare robot dexterity against a familiar human benchmark.
3) **The Cluttered Environment Picking Benchmark (CEPB) (2023)**[4]: This benchmark is aimed at cluttered and occluded picking in warehouse-like environments. Its main contribution is to provide a more realistic and difficult test environment for evaluating perception, planning, control, and grasping under heavy clutter. Rather than relying only on a small set of simple metrics, CEPB is meant to support fair comparison across systems operating in dense scenes where object visibility and physical interference are major challenges. This makes it particularly relevant for research on bin picking, clutter removal, and warehouse automation, where the environment itself is a major part of the difficulty.

IV. Methodology: Internal Grasp Formulation

This section details the mathematical framework used to compute the internal grasping force [5][6][7][8][9]. Unlike traditional external manipulation, internal grasping relies on outward expansion against the inner cavity of an object. A generalized N-finger formulation is first presented, followed by the geometric reduction for a specific symmetric two-finger effector.

A. Contact Model and Coordinate Frames

A Soft Finger Contact (SFC) model is assumed, which is appropriate for physical interactions involving compliant surfaces, such as the rubber pads on the gripper. Unlike a hard point contact, a soft finger model transmits both linear friction forces and a localized torsional friction moment about the contact normal. Let the object reference frame be denoted by $\{O\}$, located at the center of mass. For an N-finger internal grasp, a set of contact points $i \in \{1, \dots, N\}$ is defined. Let $p_i \in \mathbb{R}^3$ represent the position of the i-th contact point relative to $\{O\}$. At each contact point, a local contact frame $\{C_i\}$ is established.

B. Outward-Pointing Contact Wrenches

In traditional external grasping, the contact normal vector points inward from the gripper toward the object's interior. In contrast, an internal grasp relies on outward expansion. Therefore, the unit normal vector $z_i \in \mathbb{R}^3$ is defined such that it points outward from the central axis of the gripper, penetrating into the inner surface of the object cavity. The local rotation matrix $R_i \in \mathbb{R}^{3\times3}$ maps the local contact frame $\{C_i\}$ to the object frame $\{O\}$.

C. Internal Friction Cone Constraints

Let

$$F_{c,i} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} f_{c,i}, \quad f_{c,i} \in \mathbb{R}, \quad f_{c,i} \geq 0$$

be the generalized contact wrench performed by the i-th finger in the local frame $\{C_i\}$. $f_{c,i}$ is the magnitude of the force applied. This model takes into account the normal force to the surface of the object, tangential forces, and also torques about the contact normal due to the soft finger pad. To maintain contact without slipping or twisting, $f_{c,i}$ must reside within the Soft Finger friction cone FC_i. Because the normal is defined as pointing outward into the object wall, $f_{z,i}$ must be strictly non-negative. The internal Soft Finger friction cone constraint limits both the linear sliding and the torsional spin: $FC_i = \left\{ f_{c,i} \in \mathbb{R}^4 \mid \sqrt{f_{1,i}^2 + f_{2,i}^2} \leq \mu f_{3,i}, \quad f_{3,i} \geq 0, \quad |f_{4,i}| \leq \gamma f_{3,i} \right\}$ where μ is the static coefficient of friction and γ is the torsional friction coefficient between the rubber pad and the internal cavity surface.

D. Grasp Matrix and Equilibrium

The localized linear forces and torsional moment applied at the i-th contact point generate a combined wrench $w_i \in \mathbb{R}^6$ on the object. The local contact wrench $f_{c,i}$ is mapped to the object frame $\{O\}$ using the expanded partial grasp matrix $G_i \in \mathbb{R}^{6\times4}$. Because the localized torque $\tau_{n,i}$ acts purely about

the normal vector n_i and generates no linear force, the fourth column of the matrix directly maps this moment. The partial grasp matrix is constructed as:

$$G_i = \begin{bmatrix} R_i & 0_{3\times3} \\ [p_i]_\times R_i & R_i \end{bmatrix} F_{c,i}$$

where $[p_i]_\times$ is the skew-symmetric matrix of the position vector p_i. The global grasp matrix $G \in \mathbb{R}^{6\times4N}$ for the entire internal grasp is constructed by concatenating the partial matrices:

$$G = \begin{bmatrix} G_1 & G_2 & \dots & G_N \end{bmatrix}$$

Let $f = \begin{bmatrix} f_{c,1}^1 & f_{c,1}^2 & \dots & f_{c,1}^4 & f_{c,2}^1 & \dots & f_{c,N}^4 \end{bmatrix}^T \in \mathbb{R}^{4N}$ be the composite vector of all local contact wrenches.

For static equilibrium against an external wrench W_{ext}, the following condition must be met:

$$W_{ext} = Gf$$

subject to $f_{c,i} \in FC_i$ for all $i \in \{1, \dots, N\}$.

$$f = G^{\#}W + N(G)\gamma$$

where the first term correspond to the forces that produce movement, the second one correspond to the internal forces (forces that are part of the null space of the G matrix). $f = f_{ext} + f_{int}$

For static force closure grasping, it is necessary to find the f_{int} that compensates for the external wrench corresponding to the force of gravity.

E. Application to a Symmetric Two-Finger Gripper

While the general internal grasp is formulated for N contacts, the specific hardware implementation for this project utilizes a symmetric two-finger parallel gripper ($N = 2$). Defining the origin $\{O\}$ at the center of the internal cavity, it is assumed the gripper expands symmetrically along the y-axis. Let d be the radius of the internal cavity. The contact positions are:

$$p_1 = \begin{bmatrix} 0 \\ -d \\ h \end{bmatrix}, \quad p_2 = \begin{bmatrix} 0 \\ d \\ h \end{bmatrix}$$

The outward-pointing unit normal vectors are $n_1 = \begin{bmatrix} 0 & -1 & 0 \end{bmatrix}^T$ and $n_2 = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}^T$. By applying these symmetric conditions, the induced torques perfectly mirror each other, naturally cancelling out along the y and z axes. Let f_e be the scalar expansion force applied by the gripper. In a purely symmetric grasp ($f_{z,1} = f_{z,2} = f_e$), the primary static equilibrium condition simplifies to ensuring the outward normal forces provide enough friction to overcome the external wrench along the z-axis (e.g., gravity mg):

$$2\mu f_e \geq mg$$

F. Dynamic grasping

Transitioning from static holding to object transportation requires redefining the external wrench to account for inertial dynamics. While the static case considers only the gravitational force vector, the dynamic wrench must consider all acceleration-induced forces. Given that the manipulator executes purely linear movements in this study, the modified wrench incorporates the maximum linear acceleration of the end-effector multiplied by the object's mass for both the x and y axes. Compensating for these peak inertial forces guarantees a secure grasp under dynamic operating conditions.

V. ADMITTANCE CONTROL

To ensure reliable pick operation, particularly when grasping delicate or dimensionally varying objects, a pure position control strategy is often insufficient. To address this, an admittance control scheme was implemented directly at the gripper level.

This strategy enables the manipulator to dynamically adjust its grasping position based on the interaction forces between the gripper jaws and the target object, achieving a stable grasp. Although the *OnRobot 2FG7* gripper is equipped with an integrated force sensor in one of its fingers, this sensor was intentionally excluded from the control loop to maintain the generality of the proposed approach [10].

The interaction force is estimated to be proportional to the low-level position tracking error. The estimated force, F_{est}, at discrete time step k is calculated as:

$$F_{est}(k) = K_{stiffness} \cdot (x(k) - x_{target}(k-1))$$

Where $x(k)$ is the actual measured position of the gripper at the current time step, and $x_{target}(k-1)$ is the commanded target width from the previous control cycle. $K_{stiffness}$ is the equivalent stiffness constant of the system (virtual spring), incorporating both the mechanical compliance of the gripper mechanism and the assumed elasticity of the target object[11][12].

Once the interaction force is estimated, it is compared against F_{target}, the force target previously computed with the grasp matrix, which represents the ideal grasping force required to secure the object without causing deformation or damage. The force error is then fed into an admittance controller to generate a new target position for the low-level position control loop.

The discrete-time admittance control law is defined by the following difference equation:

$$x_{target}(k) = x_{target}(k-1) - K_{admittance} \cdot (F_{target} - F_{est}(k))$$

Where $x_{target}(k)$ is the commanded target position for the gripper jaws for the next control cycle and $K_{admittance}$ is the admittance gain, which dictates the compliance of the gripper. A higher $K_{admittance}$ results in a more compliant grasp (yielding more to force errors), while a lower value results in stiffer behaviour. This formulation allows the gripper to act as a mechanical low-pass filter.

If the estimated force $F_{est,k}$ is less than the target force F_{target}, the term $(F_{target} - F_{est,k})$ evaluates to a positive error, causing the position target $x_{target,k}$ to decrease (effectively opening the jaws further). Conversely, if the estimated force exceeds the target, the admittance law commands the jaws to close slightly, thereby reducing the applied force and preventing damage to the manipulated object.

To prevent integral windup and ensure operational safety, the admittance target is bounded by predefined maximum opening and closing hardware limits.

Furthermore, to guarantee a stable grasp before state transition (from grasping to holding), a settling evaluation mechanism was introduced. The system confirms a successful grasp only when two conditions are simultaneously satisfied over a predefined number of control cycles: the estimated force reaches the target within a defined tolerance, and the physical gripper mechanism stalls (measured by the discrete change in actual position falling below a minimum threshold). The tuning of the admittance ($K_{admittance}$) and stiffness ($K_{stiffness}$) parameters, as well as the stall thresholds, was performed empirically via a trial-and-error approach.

VI. Experimental setup

The chosen simulation environment for this study is Isaac Sim, a robotics simulation toolkit developed by NVIDIA. Isaac Sim provides real-world lighting, materials, and visuals, making it ideal for creating photorealistic and physically accurate simulations. It leverages NVIDIA's PhysX engine for real-time, high-fidelity physics simulation, which is critical for accurately modeling robotic manipulation, contact dynamics, and grasping [13].

Two types of hollow objects were designed for the experiments: one featuring a circular cavity and another with a rectangular cavity (Figure 1). At runtime, each object is dynamically assigned a randomized weight, material, and color to facilitate future implementations involving Vision-Language-Action (VLA) models. The material properties are selected from a predefined set, with their respective static friction coefficients defined as follows: wood (0.5), plastic (0.4), rubber (0.9), metal (0.25), ceramic (0.45), copper (0.3), cardboard (0.6), and glass (0.2). Furthermore, the mass of the objects is uniformly sampled from a range of 0.050 kg to 1.950 kg. During initialization, objects are spawned with random poses (position and orientation) on a workspace table situated either directly in front of or beside the robot.

The robotic platform consists of a *Universal Robots UR5e* manipulator equipped with an *OnRobot 2FG7* parallel gripper [14][15]. The manipulator is mounted on a *Mobile Industrial Robots MiR250* mobile base. However, the mobile base remains stationary during these specific experiments.

The experimental task requires the system to grasp a randomly selected object, execute a stable lift, perform linear translations in both vertical and horizontal directions, and finally release the object to let it fall.

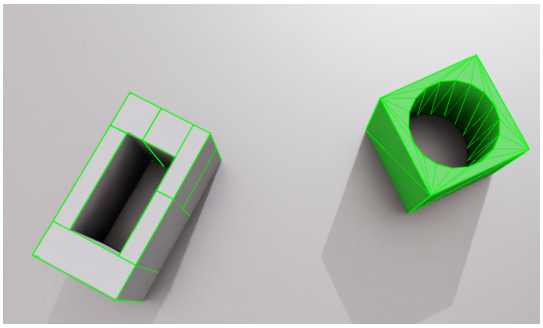


Fig. 1. Colliders of hollow objects

VII. Results

To evaluate the efficacy of the proposed mathematical framework and admittance control scheme, the testing protocol was adapted from the bin-picking benchmark proposed by Mnyusiwalla et al., chosen for its comprehensive and generalized approach.

The experimental scenarios in this study are categorized strictly by the geometry of the object's cavity: rectangular and circular. Traditional benchmark scenarios that evaluate object clutter density were intentionally omitted. Because internal grasping utilizes the internal cavity of the object, the kinematic viability of the grasp is inherently unaffected by a lack of external clearance between adjacent items (figure 2).

A trial was recorded as successful if the robot could lift the object more than 20 cm off the table and translate it without slippage.

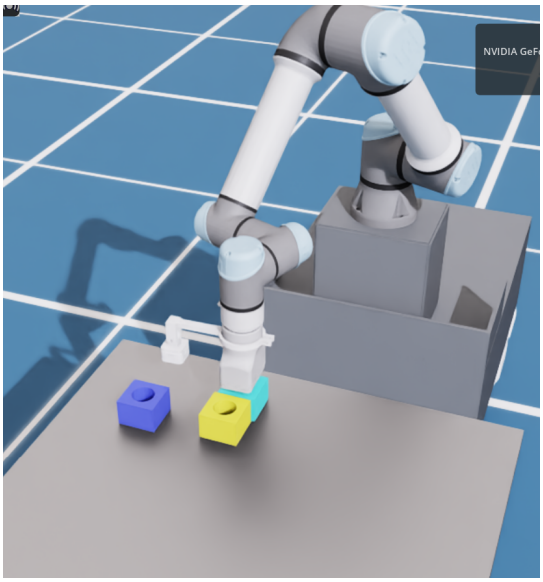


Fig. 2. Internal grasping doesn't need space around the object

A. Rectangular hole

The first set of experiments focused exclusively on grasping objects with rectangular cavities. Figure 3 presents a histogram illustrating the grasp success rate as a function of both the

object's mass and the static friction coefficient of its assigned material.

Across 100 independent simulation trials, a clear trend emerged: the success rate of the grasp decreases as the mass of the object increases. Most unsuccessful attempts in this scenario resulted from object slippage. Simulation-induced failures, such as extreme unphysical displacement (object "explosion") due to numerical instability, were observed only in exceptionally rare cases. The material properties of the object do not appear to significantly affect the overall success rate.

TABLE I
TASK SUCCESS RATE SUMMARY BY MATERIAL AND MASS GROUP FOR RECTANGULAR HOLE

Material (Coeff.)	Mass Group	Total Attempts	Successful Attempts	Success Rate
Glass (0.2)	0 – 0.5 kg	2	2	100.0%
	0.5 – 1.0 kg	4	4	100.0%
	> 1.0 kg	7	5	71.4%
Metal (0.25)	0 – 0.5 kg	3	3	100.0%
	0.5 – 1.0 kg	5	5	100.0%
	> 1.0 kg	6	4	66.7%
Copper (0.3)	0 – 0.5 kg	4	4	100.0%
	0.5 – 1.0 kg	3	1	33.3%
	> 1.0 kg	4	4	100.0%
Plastic (0.4)	0 – 0.5 kg	5	5	100.0%
	0.5 – 1.0 kg	4	4	100.0%
	> 1.0 kg	8	5	62.5%
Ceramic (0.45)	0 – 0.5 kg	4	4	100.0%
	0.5 – 1.0 kg	3	2	66.7%
	> 1.0 kg	7	6	85.7%
Wood (0.5)	0 – 0.5 kg	2	2	100.0%
	0.5 – 1.0 kg	2	2	100.0%
	> 1.0 kg	11	9	81.8%
Cardboard (0.6)	0 – 0.5 kg	2	2	100.0%
	0.5 – 1.0 kg	1	1	100.0%
	> 1.0 kg	6	3	50.0%
Rubber (0.9)	0 – 0.5 kg	2	2	100.0%
	0.5 – 1.0 kg	1	0	0.0%
	> 1.0 kg	5	4	80.0%
Total (All)	0.05 – 0.5 kg	24	24	100.00%
	0.5 – 1.0 kg	23	19	82.61%
	1.0 – 1.95 kg	54	40	74.07%

B. Circular hole

The second experimental scenario evaluated the system's performance when manipulating objects with circular cavities. Figure 4 illustrates the grasp success rate for these objects based on their material friction and mass. Consistent with the previous scenario, the grasp success rate is significantly more dependent on the object's mass than its material composition.

Unlike the rectangular objects, the primary cause of grasp failure for circular objects was not slippage, but rather severe numerical instabilities within the physics engine, resulting in the violent ejection of the object from the workspace or non-physical interpenetration between the gripper and the object. This phenomenon derives from how contact dynamics and collision meshes (colliders) are computed in Isaac Sim.

TABLE II
TASK SUCCESS RATE SUMMARY BY MATERIAL AND MASS GROUP FOR CIRCULAR HOLE

Material (Coeff.)	Mass Group	Total Attempts	Successful Attempts	Success Rate
Glass (0.2)	0 – 0.5 kg	6	6	100.0%
	0.5 – 1.0 kg	1	0	0.0%
	> 1.0 kg	2	0	0.0%
Metal (0.25)	0 – 0.5 kg	7	7	100.0%
	0.5 – 1.0 kg	5	5	100.0%
	> 1.0 kg	1	0	0.0%
Copper (0.3)	0 – 0.5 kg	4	4	100.0%
	0.5 – 1.0 kg	6	4	66.7%
	> 1.0 kg	1	0	0.0%
Plastic (0.4)	0 – 0.5 kg	7	7	100.0%
	0.5 – 1.0 kg	6	4	66.7%
	> 1.0 kg	1	1	100.0%
Ceramic (0.45)	0 – 0.5 kg	3	3	100.0%
	0.5 – 1.0 kg	6	5	83.3%
	> 1.0 kg	5	1	20.0%
Wood (0.5)	0 – 0.5 kg	3	3	100.0%
	0.5 – 1.0 kg	7	7	100.0%
	> 1.0 kg	5	2	40.0%
Cardboard (0.6)	0 – 0.5 kg	8	8	100.0%
	0.5 – 1.0 kg	4	1	25.0%
	> 1.0 kg	2	0	0.0%
Rubber (0.9)	0 – 0.5 kg	7	7	100.0%
	0.5 – 1.0 kg	5	2	40.0%
	> 1.0 kg	1	1	100.0%
Total (All)	0.05 – 0.5 kg	45	45	100.00%
	0.5 – 1.0 kg	40	28	70.00%
	1.0 – 1.95 kg	17	5	29.41%

In the simulation environment, basic primitive shapes (e.g., boxes, spheres) are strictly convex and computationally robust, but they cannot have holes. To model a rectangular cavity, the object's collider can be constructed using a compound of flat, primitive rectangular surfaces, which the physics engine handles reliably. Instead, representing a circular hole requires complex convex decomposition, resulting in a dense mesh composed of numerous small, approximated triangles (Figure 1).

During the grasping phase, Isaac Sim must resolve the intricate contact dynamics between the complex collider of the gripper and the triangular mesh of the circular inner cavity. As the applied internal radial force increases, minor mesh interpenetrations between the colliders inevitably occur. The physics engine attempts to resolve these overlaps by applying instantaneously massive restorative forces, leading to the unrealistic ejection of the object.

To mitigate this issue, a fail-safe mechanism was implemented within the control architecture. This routine continuously monitors the kinematic state of the object; if a highly unrealistic velocity spike is detected, indicative of a numerical explosion, the system automatically resets the object's pose to

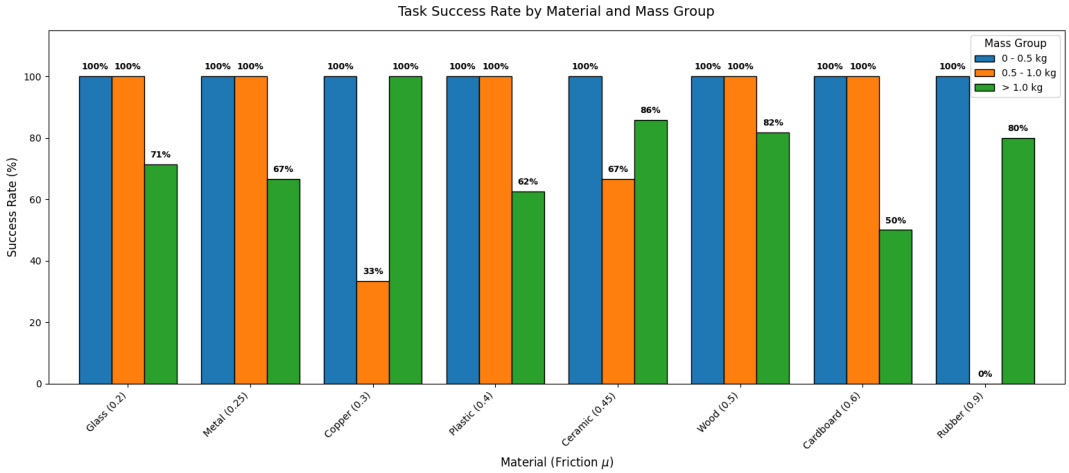


Fig. 3. Success rate for rectangular hole objects

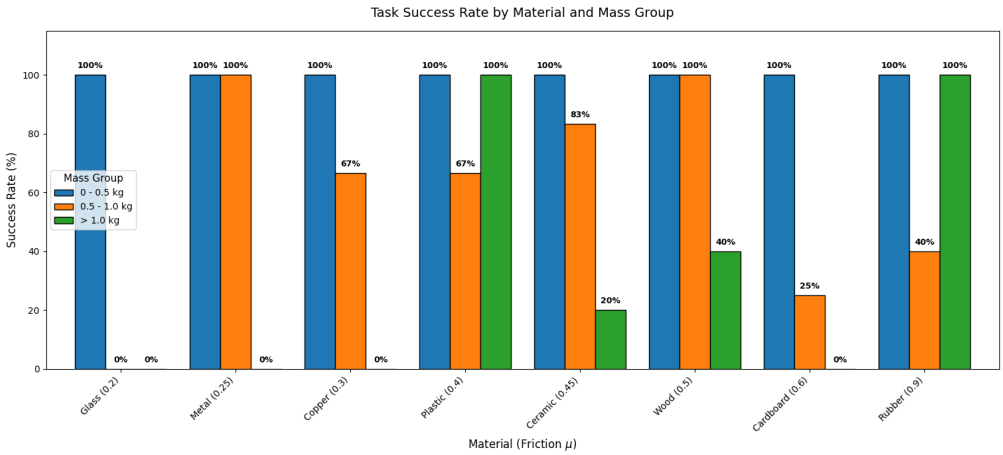


Fig. 4. Heatmap of success rate for circular hole objects

its initial state and re-attempts the grasp. However, while this mechanism reduces catastrophic simulation failures, it does not fully eliminate the underlying instability in all boundary cases, as reflected in the success rates.

C. Times

For both object geometries, the average time required to complete the full manipulation sequence, from the initial approach to the final release, ranges between 90 and 110 seconds. The time required to compute and execute the internal grasp remained consistent, varying by only a few seconds across all trials. The observed variance in total execution time was primarily driven by the object's initial random placement on the workspace table. Because this travel time is entirely dependent on the kinematic trajectory rather than the grasping framework itself, it is not considered a primary metric of interest for evaluating the proposed method's efficacy.

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